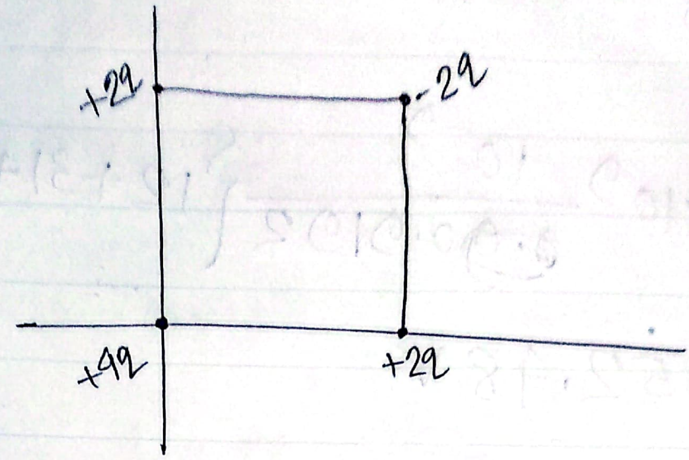
S 2

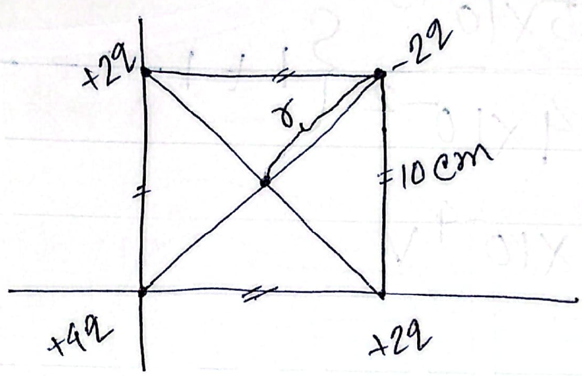
Electric Potential

Date 19-12-23

① (i)



(ii)



$r = \frac{\sqrt{2} \times 10}{2}$ cm

$= 0.07071$ m

$q = 57 \times 10^{-9}$ C

Now, $V = 9 \times 10^{9} \frac{57 \times 10^{-9}}{0.07071} [2+4+2-2]$

$= 43.53$ KV

②

(i) $V = 9 \times 10^{9} \frac{4 \times 10^{-7}}{0.09} = 40$ KV

(ii) $W = -\Delta V q = -(0 - 40,000) \times 2 \times 10^{-9}$

$= 8 \times 10^{-5}$ J.

(3) $r = \frac{1.3}{\sqrt{2}} = 0.9192\,m$

$$\therefore V = 9\times10^{9}\,\frac{10^{-9}}{0.9192}\left\{12+31+17-24\right\}$$

$$= 352.48\,V$$

(4) $$V = 9\times10^{9}\times\frac{5\times10^{-15}}{4\times10^{-2}}\left\{1+1-1-\frac{1}{2}\right\}$$

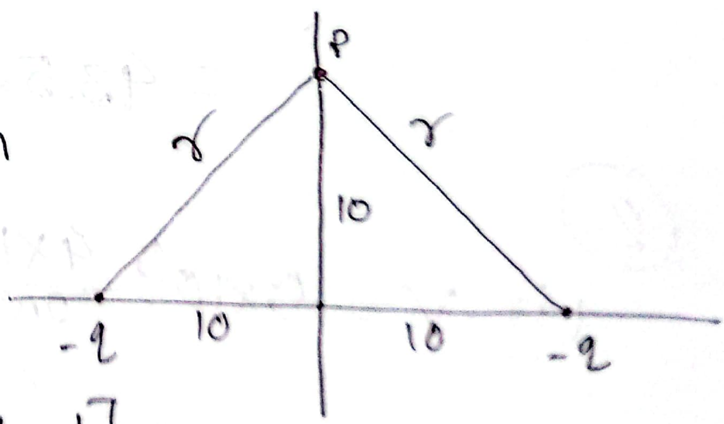
$$= 5.625\times10^{-4}\,V$$

(5) $U = -\Delta V q = 1.2\times10^{9}\times1 = 1.2\times10^{9}\,eV$

(6) Here, $r = 10\sqrt{2}$ cm

$= 0.14142\,m$

$q = 10\times10^{-5}\,c$



$$\therefore V = 9\times10^{9}\,\frac{10^{-6}}{0.14142}\left\{-1-1\right\}$$

$$= -127.28\,KV$$

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7 $V = Er$

$= 5 \times 60 \times 10^{-2} = 3V$

Now,

$3 = 9 \times 10^9 \dfrac{q}{0.6}$

$\therefore q = 2 \times 10^{-10}\, C$

9 ~~8~~

$V = 9 \times 10^9 \dfrac{p \cos\theta}{r^2}$

$= 9 \times 10^9 \dfrac{1.47 \times 3.34 \times 10^3 \times \cos 0^\circ}{(52 \times 10^{-9})^2}$

$= 1.634 \times 10^{-5}\, V$

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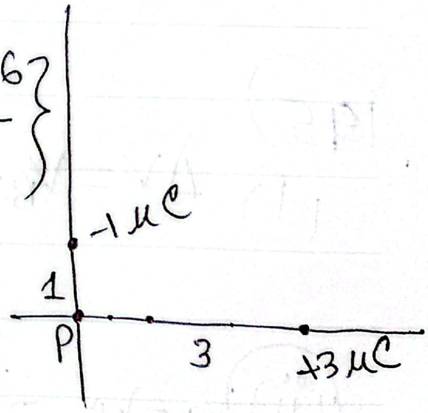
(i) $V = 9 \times 10^9 \left\{ \dfrac{-1 \times 10^{-6}}{1} + \dfrac{3 \times 10^{-6}}{3} \right\}$



$= 9 \times 10^9 \times 0$

$= 0V$

(ii) $U = -\Delta V q$

$= -(V_i - V_f) q$

$= -(0 + 0) \times 3 \times 10^{-6} = 0J$

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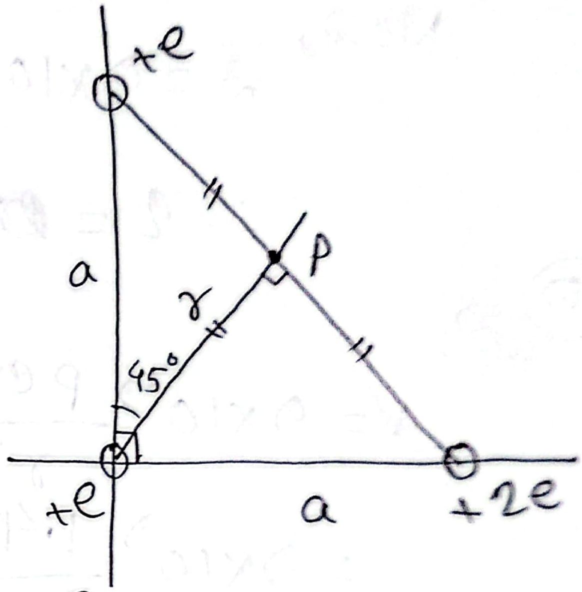
$$V = 9\times10^{9}\,\frac{1.6\times10^{-19}}{2\times10^{-2}}\{1-1+1+1-1\}$$

$$= 7.2\times10^{-8}\,V$$

14 $r = a\cos 45^\circ = \frac{a}{\sqrt{2}}$

$\therefore r = 4.24269\times10^{-6}\,m$



$$\therefore V = 9\times10^{9}\,\frac{1.6\times10^{-19}}{4.24269\times10^{-6}}\{1+1+2\}$$

$$= 1.36\times10^{-3}\,V$$

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(i) $\Delta V = V_B - V_A = \frac{-W}{q} = \frac{-3.94\times10^{-19}}{-2\times1.6\times10^{-19}}$

$= 1.23\,V$

(ii) $V_C - V_A = 1.23\,V$

we know,

$W = qEd\cos\theta$

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From figure, for A to c

distance $= \frac{d}{\cos\theta}$

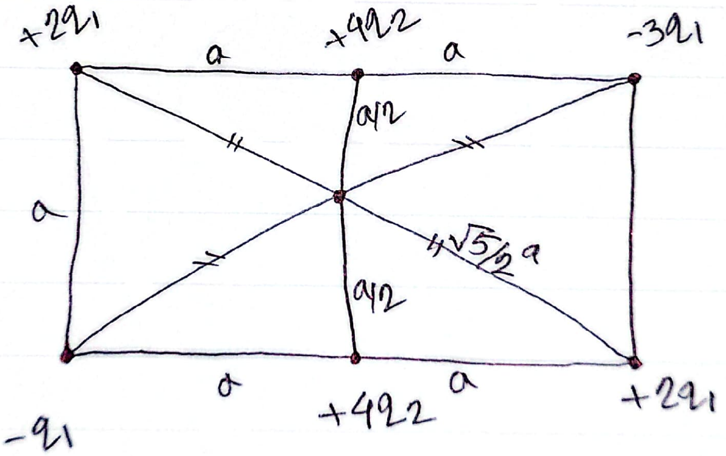
$\therefore \omega = qE \frac{d}{\cos\theta} \cos\theta$

$= qEd$

confused about this explanation.

(iii) $V_C - V_B = 0$ [E and d are perpendicular]

(16)



$\sqrt{a^2+\left(\frac{a}{2}\right)^2}$

$= \frac{\sqrt{5}}{2} a$

$q_1 = 3.4 \times 10^{-12} C$

$q_2 = 6 \times 10^{-12} C$

$a = 0.39 m$

$\therefore V = 9 \times 10^9 \times \frac{1}{a/2} \left\{ \frac{2q_1}{\sqrt{5}} - \frac{q_1}{\sqrt{5}} + 4q_2 + 4q_2 - \frac{3q_1}{\sqrt{5}} + \frac{2q_1}{\sqrt{5}} \right\}$

$= 2.215 V$

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$$V = 9\times10^{9}\,\frac{1.6\times10^{-19}}{1.4\times10^{-2}}\left\{\frac{4}{2}-\frac{4}{3}\right\}$$

$$= 6.857\times10^{-8}\,V$$

$$\omega = 6.857\times10^{-8}\times16\times1.6\times10^{-19}$$

$$= 1.7555\times10^{-25}\,J$$